

DARIUS-ANDREI BOBELNICEANU

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Education

Bachelor's Degree in Computer Science and Engineering

Oct. 2022 – July 2026

Faculty of Automatic Control and Computer Science, University Politehnica of Bucharest

Relevant Coursework

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|----------------------------------|-------------------------------|---------------------|---------------------------------------|
| • Computers Programming | • Object Oriented Programming | • Operating Systems | • Parallel and Distributed Algorithms |
| • Data Structures and Algorithms | • Analysis of algorithms | • Database | • Networks |
| | | • Computer Graphics | |

Projects

🔗 IMDb App | Java

- Developed an IMDb 2.0 application using **Java** to create an intuitive and user-friendly interface, **enhancing the overall user experience**.
- Improved code **structure, reusability, and maintainability** by applying **design patterns**, such as Singleton, Builder, Factory, Observer, and Strategy, ensuring **efficient maintenance of the application**.
- Integrated robust functionalities including movie/series/actor management, search capabilities, a request system for user interaction and a notification system utilizing the Observer pattern to provide **accurate and timely updates**.

🔗 Libra Connect | C

- Developed a **command-line client application** that communicates with a **RESTful API** to perform **CRUD** operations on a digital library system, including **user registration, login, and book management**.
- Implemented **secure token-based authentication using cookies and JWTs**, enabling users to access protected routes like the library and specific book details.
- Designed and handled low-level socket programming for **TCP/IP communication, parsing HTTP responses**, ensuring robust error handling and input validation.

🔗 Image Compressor | C

- Used a quadtree to compress images** with a given factor by recursively dividing the image into four subspaces with each holding the average RGB color and the error determining that color for its subspaces.
- Implemented an application which can compress and decompress any given image in PPM format. After applying compression, **the image size was reduced by up to 98%**.

🔗 Artillery Combat Game Engine | OpenGL/C++

- Developed a **physics-based 2D combat game engine** featuring real-time terrain deformation, dynamic collision detection, and ballistic trajectory simulation.
- Implemented **advanced graphics** rendering using OpenGL, including custom mesh generation, matrix transformations, and dynamic terrain modification.
- Designed modular, **object-oriented architecture** allowing for easy expansion of game features and mechanics.

Extracurricular

Logistics Department Manager

2024

"LSAC" - Computer Science Students' League

- Successfully lead a team of **30 people**, ensuring all logistical tasks were **executed efficiently** and on schedule.
- Worked in **strategic partnership** with the event's core organizational team, driving **crucial initiatives** that elevated the event's overall **impact** and significantly boosted attendee engagement to 1.000+ individuals, that enriched the overall experience.

Leadership Development Program

2025

powered by Sievo and Leaders Foundation

- Completed an intensive curriculum focused on strategic **project management, decision-making, adaptive leadership**, emotional intelligence, and entrepreneurial thinking.
- Collaborated in interdisciplinary teams to develop and pitch solutions to **real-world challenges**, demonstrating leadership in **high-pressure scenarios** and delivering **measurable impact**, guided by established industry leaders.

Robotics "RoboTitans" team

2020 – 2021

- Participant in the FIRST Tech Challenge as a member of the RoboTitans robotics team, in which our team was awarded the "Think Award" for our **innovative solutions and exceptional performance**.
- Part of the 3D Design and Construction department, utilizing the Onshape application.

Technical Skills

Programming Languages: C, C++, Java, Linux, Shell Scripting, Python, SQL, HTML, CSS

Languages: English (C1 - Cambridge), Romanian (Native)